# Lecture 6 Compendium: Remote Sensing Strategies in Agriculture

Module 2: Technologies of IoT in Crop Production

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Compendium for Advanced Study

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# 1 The Remote Sensing Continuum

Remote sensing is the art and science of obtaining information about an object without being in physical contact with it. In agriculture, we operate across a **Continuum of Scales**.

| Platform | Type | Altitude | Spatial Res | Temporal Res |
|---|---|---|---|---|
| Satellite | Macro | 500 - 800 km | 10 - 30 m | 3 - 16 Days |
| Aerial (UAV) | Meso | 50 - 120 m | 1 - 5 cm | On-Demand |
| Proximal (Ground) | Micro | 0.5 - 2 m | $<$ 1 cm (Point) | Real-Time |

Table 1: The Scales of Agricultural Remote Sensing

**The PhD Challenge**

**The Mixed Pixel Problem:** A 10m Satellite pixel (Sentinel-2) covering a smallholder farm in Tanzania might contain Maize, Beans, Soil, and a House. De-convoluting this signal (Spectral Unmixing) is a key research area.

# 2 Physics of Radiometry: Spectral Signatures

To interpret satellite or drone images, we must understand how electromagnetic radiation interacts with leaf tissue.

## 2.1 Interaction Mechanisms

When light hits a leaf, it is either: 1. **Absorbed ($\alpha$):** Energy taken in (mostly for Photosynthesis). 2. **Transmitted ($\tau$):** Energy passing through. 3. **Reflected ($\rho$):** Energy bouncing back (This is what sensors see).

$$\alpha + \tau + \rho = 1$$

## 2.2 The Vegetation Curve

Healthy green vegetation has a distinct spectral signature:

- **Visible Blue (450nm) & Red (670nm):** High Absorption by Chlorophyll a/b. Low Reflectance ($\rho \approx 0.1$).
- **Visible Green (550nm):** Moderate Reflectance (hence plants look green).
- **Red Edge (700-740nm):** The transition zone. Slope indicates stress.
- **Near Infrared (NIR, 750-1300nm):** Very High Reflectance ($\rho > 0.5$). Caused by internal scattering in the spongy mesophyll structure.
- **Short Wave Infrared (SWIR, 1400-2500nm):** Absorption by Water content.

**Research Implication:** Stressed plants break down chlorophyll (Red reflectance increases) and their cell structure collapses (NIR reflectance decreases).

# 3 Vegetation Indices: The Mathematics of Health

We combine bands mathematically to isolate specific plant traits and remove background noise (soil/atmosphere).

### 3.1 Normalized Difference Vegetation Index (NDVI)

The standard bearer. Measures "Greenness" or Biomass.

$$NDVI = \frac{\rho_{NIR} - \rho_{Red}}{\rho_{NIR} + \rho_{Red}} \tag{1}$$

* **Range:** -1 to +1. * **Typical Values:** 0.2 (Soil), 0.5 (Sparse Crop), 0.8 (Dense Canopy). * **Saturation Issue:** At high biomass (Leaf Area Index $> 3$), NDVI saturates and cannot distinguish between dense and very dense canopies.

### 3.2 Soil Adjusted Vegetation Index (SAVI)

Corrects for soil background noise in sparse crops (early growth stage).

$$SAVI = \frac{(\rho_{NIR} - \rho_{Red}) \cdot (1 + L)}{\rho_{NIR} + \rho_{Red} + L} \tag{2}$$

Where $L$ is the soil adjustment factor (usually 0.5).

### 3.3 Normalized Difference Red Edge (NDRE)

Uses the Red Edge band instead of Red. It penetrates deeper into the canopy and does not saturate as quickly as NDVI. Critical for late-stage Nitrogen monitoring.

$$NDRE = \frac{\rho_{NIR} - \rho_{RedEdge}}{\rho_{NIR} + \rho_{RedEdge}} \tag{3}$$

### 3.4 Python: Calculating Indices from Multispectral Data

Simulating the calculation of these indices from a 5-band sensor (Blue, Green, Red, RedEdge, NIR).

```python
import numpy as np

def calculate_indices(red_band, nir_band, re_band, soil_factor=0.5):
    """
    Calculate NDVI, SAVI, and NDRE.
    Input bands should be float reflectance values (0.0 to 1.0).
    """
    # Avoid division by zero
    epsilon = 1e-10

    # 1. NDVI
    ndvi_num = (nir_band - red_band)
    ndvi_den = (nir_band + red_band) + epsilon
    ndvi = ndvi_num / ndvi_den

    # 2. SAVI
    savi_num = (nir_band - red_band) * (1 + soil_factor)
    savi_den = (nir_band + red_band + soil_factor) + epsilon
    savi = savi_num / savi_den

    # 3. NDRE
    ndre_num = (nir_band - re_band)
    ndre_den = (nir_band + re_band) + epsilon
    ndre = ndre_num / ndre_den

    return ndvi, savi, ndre

# Example Data (Pixel values)
```

```
# Healthy Plant: Low Red, High NIR
pixel_healthy = {'red': 0.05, 'nir': 0.60, 're': 0.20}
# Stressed Plant: Higher Red, Lower NIR
pixel_stress =  {'red': 0.15, 'nir': 0.35, 're': 0.25}

h_ndvi, h_savi, h_ndre = calculate_indices(pixel_healthy['red'], pixel_healthy['
    nir'], pixel_healthy['re'])
s_ndvi, s_savi, s_ndre = calculate_indices(pixel_stress['red'], pixel_stress['
    nir'], pixel_stress['re'])

print(f"--- Healthy Plant ---")
print(f"NDVI: {h_ndvi:.2f} | SAVI: {h_savi:.2f} | NDRE: {h_ndre:.2f}")
print(f"--- Stressed Plant ---")
print(f"NDVI: {s_ndvi:.2f} | SAVI: {s_savi:.2f} | NDRE: {s_ndre:.2f}")
```

Listing 1: Calculating Vegetation Indices

# 4 Active vs. Passive Sensing

This distinction dictates the operational window of your research.

## 4.1 Passive Sensing (The Sun)

Reflects solar energy.

- **Examples:** Satellites (Sentinel-2, Landsat), Drones with Cameras.
- **Pros:** Rich spectral information (Hyperspectral).
- **Cons:** Requires sunlight. Cannot work at night. Affected by shadows and clouds. **Data must be radiometrically calibrated** (Atmospheric Correction).

## 4.2 Active Sensing (The Device)

Emits its own energy pulse and measures the return.

- **Examples:** LiDAR (Laser), Radar (SAR), GreenSeeker (LED).
- **Pros:** Works day/night. Penetrates clouds (Radar). Inherently calibrated.
- **Cons:** Limited spectral bands (usually single wavelength). Power hungry.

**Proximal Sensing**

**GreenSeeker:** A handheld active sensor. It fires Red and NIR LEDs at the plant and measures reflectance. It calculates NDVI instantly, regardless of cloud cover. This is the standard for ground-truthing satellite data.

# 5 Advanced Topics: Synthetic Aperture Radar (SAR)

For tropical regions like Tanzania, clouds block optical satellites for 6 months a year (Rainy Season). **Radar (SAR)** is the solution.

### 5.1 Physics of Backscatter

SAR (Sentinel-1) transmits Microwave pulses (C-band, 5cm wavelength). * **Roughness:** Rough surfaces (Leaves) scatter energy back (High Backscatter). Smooth surfaces (Water, Bare Soil) reflect energy away (Low Backscatter). * **Dielectric:** Wet objects reflect more radar energy.

**Application:** 1. **Rice Monitoring:** Flooded paddies look black (Specular reflection). As rice grows, the signal becomes brighter (Volume scattering). 2. **Soil Moisture:** In bare fields, backscatter intensity correlates linearly with surface soil moisture.

## 6 Review Questions & Research Assignment

1. **Physics:** Explain why a plant suffering from water stress might not show a change in NDVI immediately, but would show a change in NDWI (Normalized Difference Water Index) or Thermal IR.

2. **Math:** Calculate the NDVI for a pixel with Red Reflectance = 0.1 and NIR Reflectance = 0.5. If atmospheric haze increases the Red reading by 0.05 but leaves NIR unchanged, what is the

3. **Design:** You are monitoring a Rice Paddy in Kilombero during the heavy rains. Optical satellites show nothing but clouds. Design a monitoring strategy using Sentinel-1 (Radar) to estimate the growth stage of the rice.

## References

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3. Trimble Agriculture. *GreenSeeker Handheld Crop Sensor User Guide*.